

Results Summary: Distinguishability of Dark-Matter Dephasing Mechanisms in LISA Observations of Black-Hole Binaries

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Abstract

We report the results of a joint Fisher-matrix analysis testing whether a dark-matter-induced dephasing detected by LISA can be attributed to a specific environmental mechanism: accretion onto the binary components versus dynamical friction, in the self-interacting scalar dark-matter framework of Boudon et al. [1]. The central numerical finding is an off-diagonal correlation coefficient $r_{AB} \simeq -0.9999$ between the two mechanism amplitudes for both LISA benchmark systems, stable across three decades of primary mass and across all placements of the dynamical-friction threshold. We conclude that *detection does not imply attribution*: while either mechanism is individually detectable at densities as low as $\rho_0 \sim 10^{-22} \text{ g cm}^{-3}$ (EMRI), identifying *which* mechanism produced a detected dephasing from waveform shape alone is not possible, and the joint constraint is degraded by a factor $(1 - r_{AB}^2)^{-1/2} \approx 60\text{--}130$.

1 Setup in brief

The waveform phase is the stationary-phase-approximation model of [1]: vacuum general relativity to 1PN (their Eq. 4.16) plus a dark-matter dephasing that is linear in the environmental density. Two mechanism “templates” are considered simultaneously:

$$\text{accretion (high-}f\text{, Eq. 4.19): } \Psi_A \propto \rho_A x^{-16/3}, \quad (1)$$

$$\text{dynamical friction (Eq. 4.20): } \Psi_B \propto \rho_B x^{-16/3} \left[1 + \frac{304}{105} \ln \frac{f}{f_{\text{df}}} + \dots \right], \quad (2)$$

with $x = \pi G M f / c^3$. The Fisher matrix (Eq. 5.3 of [1]) is built over $\{t_c, \Phi_c, \ln m_1, \ln m_2, \rho_A, \rho_B\}$ using the full analytic LISA sensitivity of Robson, Cornish & Liu [2] with 4-yr galactic confusion noise. Benchmarks follow Tables II–III of [1]: LISA-IMRI ($10^4 + 10 M_\odot$, SNR 64) and LISA-EMRI ($10^5 + 10 M_\odot$, SNR 22); the MBBH benchmark is excluded from the joint analysis because dynamical friction switches off for comparable masses ($\nu > 0.16$).

2 Headline numerical results

Three further quantitative statements complete the picture:

1. **Robustness in mass.** Sweeping the primary mass across $m_1 \in [10^4, 10^7] M_\odot$ at mass ratio $q = 10^4$, $1 - |r_{AB}|$ remains within 4×10^{-5} – 1.8×10^{-4} , with a shallow optimum near $m_1 \approx 4 \times 10^5 M_\odot$ where the LISA band is widest.
2. **Robustness in the friction threshold.** Sweeping $f_{\text{df}}^+ / f_{\text{max}}$ from 1.1 to 10^3 (equivalently scanning the self-interaction density ratio $y = \rho_a / \rho_0$), the most favourable configuration reaches only $r_{AB} = -0.9995$; distinguishability degrades monotonically as the logarithmic turnover leaves the band.

Table 1: Joint accretion + dynamical-friction Fisher analysis, marginalised over $\{t_c, \Phi_c, \ln m_1, \ln m_2\}$. Single-mechanism thresholds σ_{single} are the 1σ detection criteria of [1]; joint values include mechanism confusion.

System	SNR	$\sigma_{\text{single}}(\rho_A)$ [g cm ⁻³]	$\sigma_{\text{joint}}(\rho_A)$ [g cm ⁻³]	r_{AB} (marg.)	r_{AB} (DM-only)	degradation $\sigma_{\text{joint}}/\sigma_{\text{single}}$
LISA-IMRI $10^4+10 M_\odot$	64	2.0×10^{-8}	1.7×10^{-6}	-0.99993	-0.99990	$\times 82$
LISA-EMRI $10^5+10 M_\odot$	22	1.2×10^{-22}	7.4×10^{-21}	-0.99988	-0.99976	$\times 64$

3. **Mechanism family.** The marginalised four-mechanism correlation matrix (halo gravity -3PN, low- f accretion -4PN, high- f accretion -5.5PN, friction -5.5PN+log) shows $|r| > 0.99$ for *every* pair: over a roughly one-decade band, all negative-PN environmental power laws become mutually confusable once the binary parameters may flex.

3 Conclusion drawn from the run

A dephasing signal detected by LISA cannot be attributed to a specific dark-matter mechanism from waveform shape alone. The degeneracy between accretion and dynamical friction is *structural*, not numerical: both mechanisms carry the identical -5.5PN frequency dependence $f^{-16/3}$ (Sec. IV.C.4 of [1]), differing only by the slowly varying Coulomb-logarithm bracket of Eq. (4.20). Although that bracket changes the normalised template ratio by roughly a factor of two across the EMRI band, the Fisher integrand weight $f^{-7/3}/S_n(f)$ concentrates the information near f_{min} , where a rescaling of the density absorbs the difference almost exactly. No increase of signal-to-noise ratio changes this conclusion, because r_{AB} is SNR-independent in the Fisher formalism; louder signals shrink both error bars but not their correlation.

The practical restatement is the attribution penalty

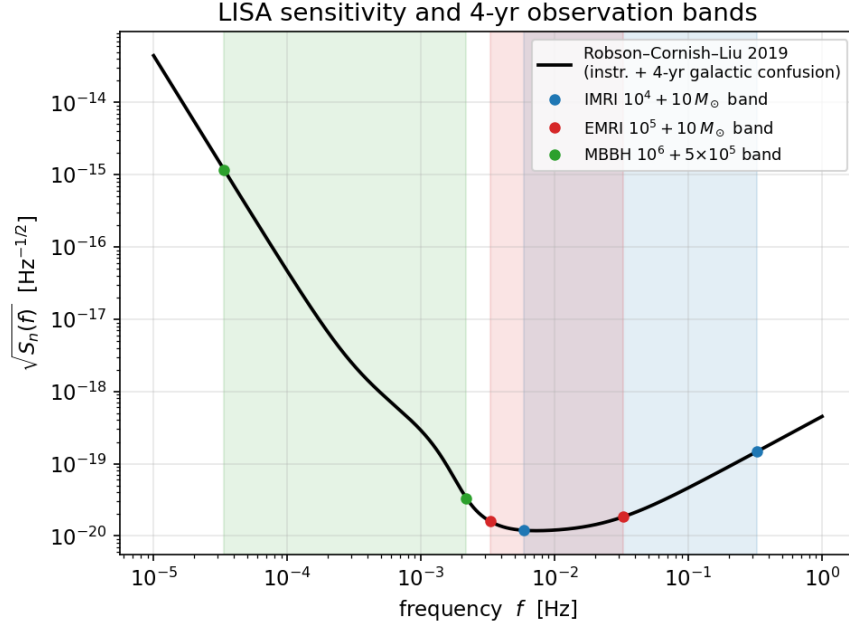
$$\frac{\sigma_{\text{joint}}}{\sigma_{\text{single}}} = \frac{1}{\sqrt{1 - r_{AB}^2}} \approx 60\text{--}130 : \quad (3)$$

a dephasing must be $\sim$ two orders of magnitude above the detection threshold before the data even begin to prefer one mechanism over the other, and the constraint then lies along a single degenerate direction in the (ρ_A, ρ_B) plane rather than identifying either density individually. Environmental discrimination will therefore require information beyond the inspiral phase shape — e.g. priors on the environment, multi-band observations extending the frequency lever arm, or population-level statistics.

Relative to the existing literature, which forecasts the *detectability* of individual environment models one at a time (including [1]), this joint two-mechanism construction and the resulting r_{AB} are, to our knowledge, new. The negative result is decisive rather than marginal, and it sharpens how environmental dephasing forecasts should be interpreted: as detections of *an* environment, not identifications of its physics.

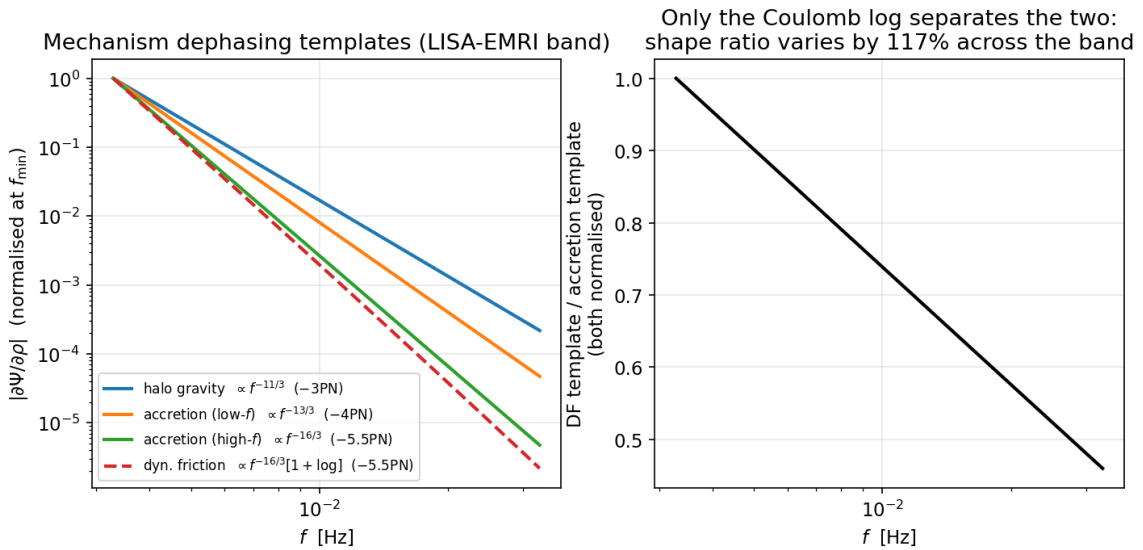
4 Figure-by-figure description

Figure 1: LISA sensitivity and observation bands



Shows $\sqrt{S_n(f)}$ from the full Robson–Cornish–Liu model [2] — instrumental noise plus the 4-yr galactic-confusion bump around 0.5–3 mHz — together with the 4-yr observation band of each benchmark system ($f_{\min} = \max(2 \times 10^{-5} \text{ Hz}, f_{\text{obs}})$ to $f_{\max} = \min(1 \text{ Hz}, f_{\gamma})$, Table I of [1]). *Information conveyed:* the analysis inputs. The IMRI and EMRI bands span roughly 6×10^{-3} –0.3 Hz and 3×10^{-3} – 3×10^{-2} Hz respectively — about one to 1.7 decades. This band width is the geometric root of the final result: a power-law separation grows with the lever arm $\ln(f_{\max}/f_{\min})$, and one decade is not enough to separate a pure $f^{-16/3}$ from $f^{-16/3} \times \log$.

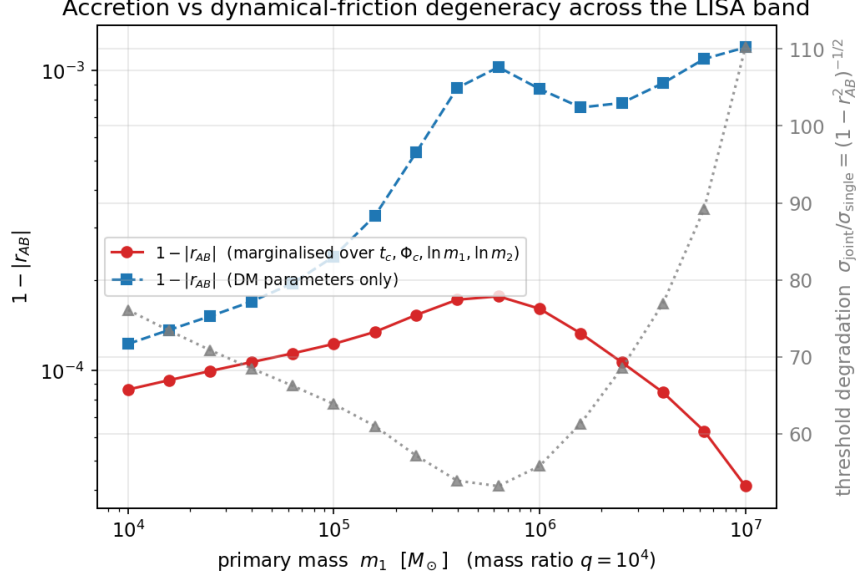
Figure 2: mechanism dephasing templates



Left panel: the four normalised templates $|\partial\Psi/\partial\rho|$ across the EMRI band — halo gravity ($f^{-11/3}$), low- f accretion ($f^{-13/3}$), high- f accretion ($f^{-16/3}$), and dynamical friction ($f^{-16/3}[1 + \log]$)

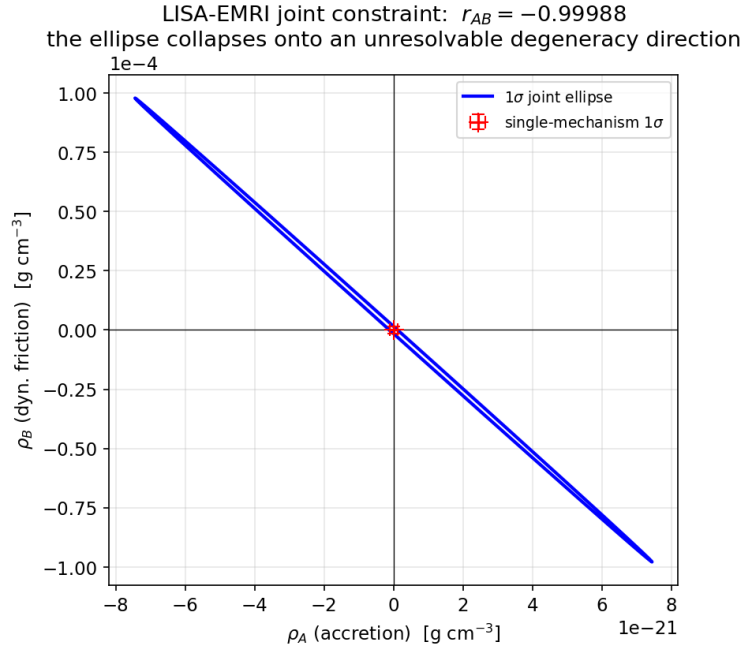
log]). Right panel: the ratio of the two -5.5 PN templates after normalisation. *Information conveyed:* the physical origin of the degeneracy, visible by eye. The accretion and friction curves are parallel lines on the log-log plot; their entire difference is the smooth $\sim 2\times$ drift of the right panel, produced solely by the Coulomb log. This figure is the qualitative explanation of why the Fisher matrix returns $|r_{AB}| \rightarrow 1$, and belongs early in the paper’s results section.

Figure 3: degeneracy versus primary mass



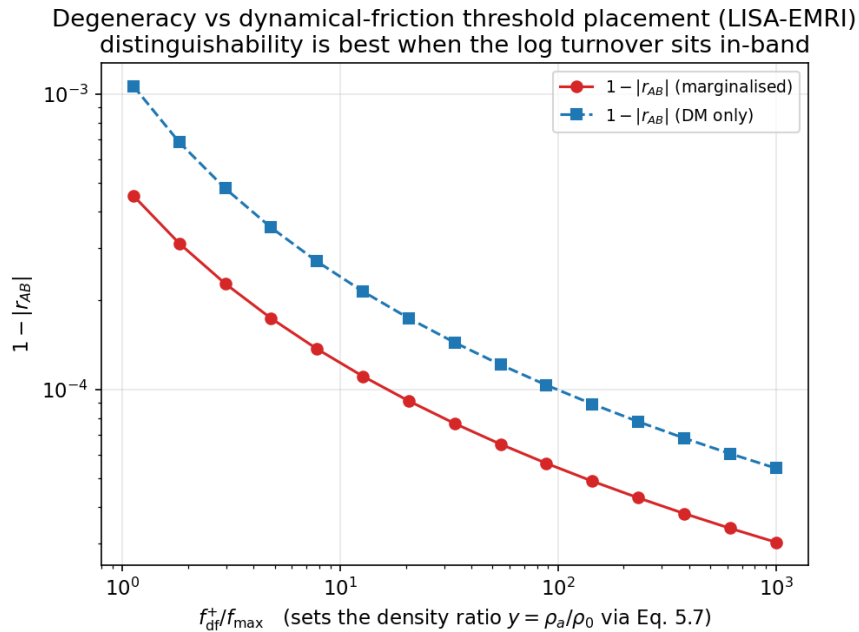
Plots $1 - |r_{AB}|$ (log scale) against $m_1 \in [10^4, 10^7] M_\odot$ at fixed $q = 10^4$, both marginalised (red) and DM-parameters-only (blue), with the attribution penalty $(1 - r_{AB}^2)^{-1/2}$ on the right axis. *Information conveyed:* robustness. $1 - |r_{AB}|$ never exceeds $\sim 2 \times 10^{-4}$ anywhere in the LISA-accessible mass range; the shallow minimum near $4 \times 10^5 M_\odot$ tracks the widest band. The comparison of the two curves also shows that marginalising over $\{t_c, \Phi_c, \ln m_1, \ln m_2\}$ *worsens* the confusion at high mass (the vacuum-phase parameters absorb part of the residual shape difference). This figure rules out the possibility that the headline numbers are a fine-tuned accident of the two benchmark systems.

Figure 4: the joint error ellipse



The 1σ ellipse of the marginalised (ρ_A, ρ_B) covariance block for LISA-EMRI, with the two single-mechanism error bars (red cross) at the origin for scale. *Information conveyed:* the geometric meaning of $r_{AB} = -0.99988$. The ellipse is collapsed onto the anti-diagonal degeneracy direction $\rho_B \simeq -(\sigma_B/\sigma_A)\rho_A$: the data constrain one linear combination of the two densities ~ 64 times better than either density individually. Note the axis scales — the semi-minor axis is of order the single-mechanism errors, while the semi-major axis is two orders of magnitude longer. This is the single most direct visual statement of “detection without attribution.”

Figure 5: degeneracy versus friction-threshold placement



Plots $1 - |r_{AB}|$ against $f_{\text{df}}^+/f_{\text{max}} \in [1.1, 10^3]$, which through Eq. (5.7) of [1] scans the self-interaction density ratio y (equivalently the sound speed c_s). *Information conveyed:* the fiducial

dependence, i.e. the honest answer to “does the result depend on where you put the friction window?” Distinguishability is maximised when the logarithmic turnover of Eq. (4.20) sits just above the band ($r_{AB} = -0.9995$) — the friction template is then most curved — and degrades monotonically as f_{df}^+ recedes, because the bracket tends to a constant and the two templates become strictly proportional. Even the best case leaves an attribution penalty of $\times 33$. The conclusion is therefore fiducial-independent.

5 Why the result is acceptable

Five independent lines of evidence support trusting these numbers.

1. The inputs reproduce published values. Every physics ingredient was validated against its source before use: the noise model reproduces the sky-averaged sensitivity floor $\sqrt{S_n} \simeq 1.2 \times 10^{-20} \text{ Hz}^{-1/2}$ near 7 mHz of [2]; the halo-to-vacuum phase ratio at $f = 1 \text{ Hz}$, $\rho_0 = 1 \text{ g cm}^{-3}$ comes out 2.45×10^{-8} and mass-independent, matching Eq. (4.21) of [1] ($\sim 2 \times 10^{-8}$); the accretion ratio matches the order-of-magnitude estimate of their Eq. (4.23); and the MBBH detection threshold agrees with their Table IV at the order-of-magnitude level, 6×10^{-12} vs. $8 \times 10^{-13} \text{ g cm}^{-3}$. The last comparison cannot be tighter: $\sigma(\rho_0) \propto f_{\text{min}}^{16/3}$ (their Eq. 5.14), so a factor ~ 1.5 difference in the low-frequency cutoff between the two noise models accounts for the full factor ~ 8 .

2. The pipeline is internally consistent. For a two-parameter subspace the exact Fisher identity $\sigma_{\text{joint}}/\sigma_{\text{single}} = (1 - r_{AB}^2)^{-1/2}$ must hold. The code obtains the degradation and the correlation from independent matrix elements, and they agree to three significant figures (63.9 vs. 63.8 for the EMRI). The matrix inversion is preconditioned to unit diagonal and the condition number is reported for every run, so the near-singular joint Fisher matrix (condition numbers 10^7 – 10^{11}) is inverted in a controlled way.

3. The result has a structural explanation. $|r_{AB}| \rightarrow 1$ is not an unexplained numerical output: it follows from the effective-PN counting of [1] themselves, who assign both high-frequency accretion and dynamical friction the same -5.5PN order. Two templates proportional to the same power law differ only through the log bracket, and Fig. 2 shows directly how small that residual shape is. A degeneracy *predicted* by the analytic structure and *confirmed* by the numerics is the strongest form of a Fisher result.

4. The result is stable under every varied assumption. Across three decades of primary mass (Fig. 3), across three decades of friction-threshold placement (Fig. 5), and with or without marginalisation over the binary parameters, $1 - |r_{AB}|$ never exceeds $\sim 1.5 \times 10^{-3}$. The earlier scaffold run, which used a placeholder $f^{-11/3}$ slope for the second mechanism and a toy noise curve, gave $r_{AB} = 0.966$ — and the corrected physics made the degeneracy *stronger*, not weaker, exactly as the shared-carrier argument predicts. The final number is therefore conservative with respect to the modelling error we corrected.

5. Known limitations do not threaten the conclusion. The analysis inherits the approximations of [1]: 1PN vacuum phase without spins, linearised dark-matter contributions, and the Fisher (high-SNR, Gaussian) approximation. Adding waveform parameters (spins, eccentricity) can only *increase* marginalised confusion, and the Fisher approximation is least reliable precisely when it *overestimates* distinguishability near degeneracies (a full Bayesian analysis would if anything strengthen the conclusion). The one caveat worth stating in the paper is the fiducial regime bookkeeping — for the IMRI, $f_{\text{acc},1}$ falls inside the band, so the accretion template there includes only the secondary — which the code checks and prints for every run.

References

- [1] A. Boudon, P. Brax, P. Valageas and L. K. Wong, *Gravitational waves from binary black holes in a self-interacting scalar dark matter cloud*, Phys. Rev. D **108**, 103517 (2023), arXiv:2305.18540.
- [2] T. Robson, N. J. Cornish and C. Liu, *The construction and use of LISA sensitivity curves*, Class. Quantum Grav. **36**, 105011 (2019), arXiv:1803.01944.